Drawings

Three-body Arch from Figure 12.27

Contact (x, y) locations and centers of each masses are as annotated in the figure below

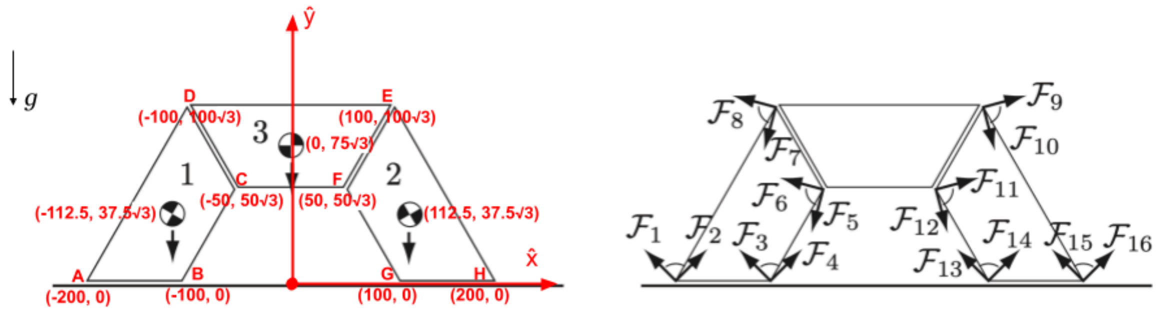

Figure 12.27: (Left) An arch under gravity. (Right) The friction cones at the contacts of stone 1 and the contacts of stone 2.

Figure 1

Graphical Explanation of Three-body Arch Continue to stand

In Left of Figure 2-4, the contact wrenches and gravity wrench were drawn when $\mu = 0.5$ and $m = 10kg$ for each body.

In the Right of Figure 2-4, all the contact wrenches were sliding along their lines of action till bases of the arrows are coincident in the center of mass. The contact wrenches, when summed together, balance against gravity wrench.

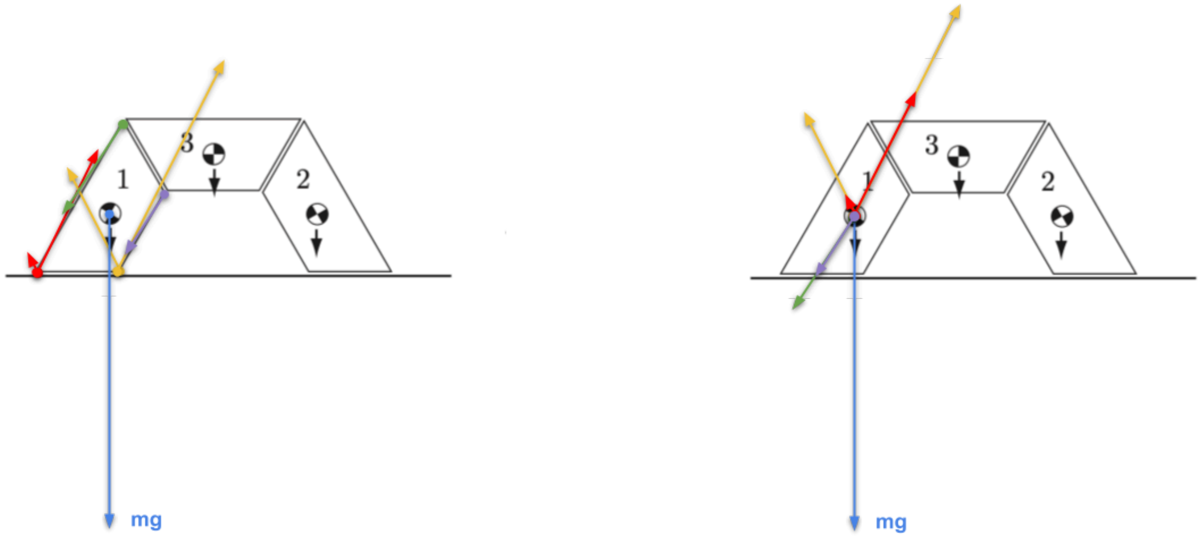

Figure 2

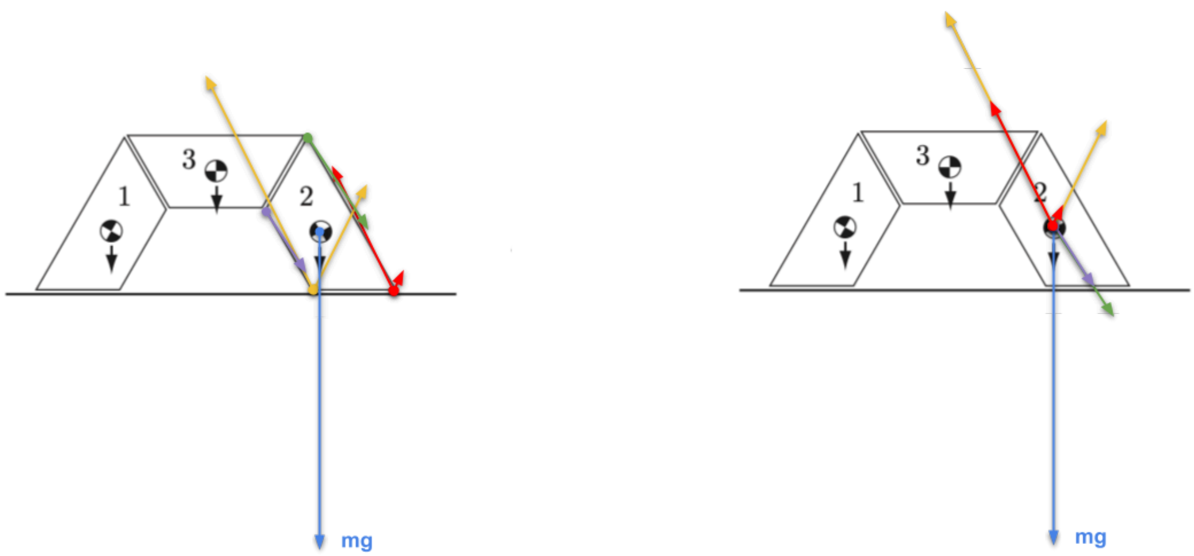

Figure 3

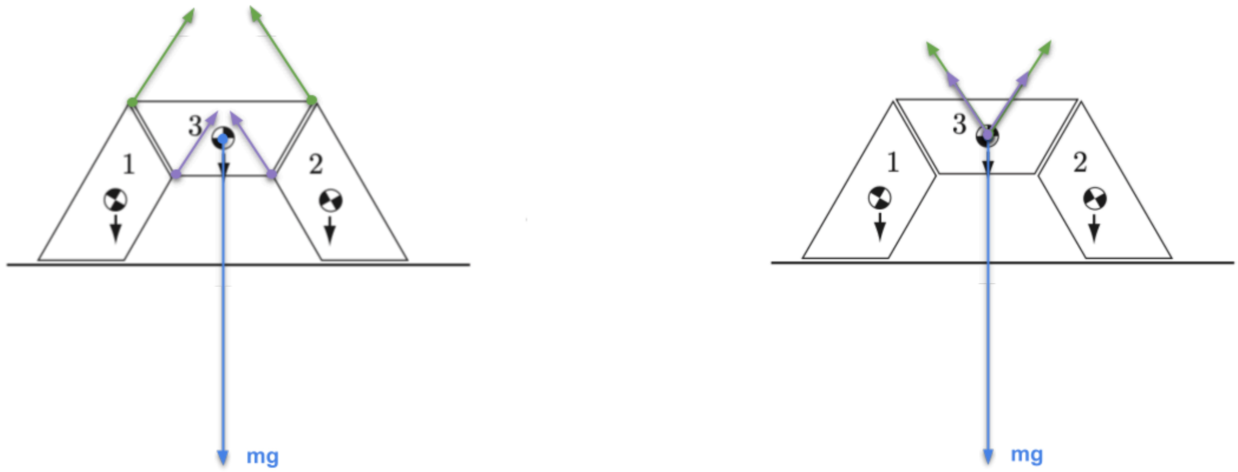

Figure 4

Graphical Explanation of Three-body Arch Collapse

Figure 5-7 show the composite wrench cone for each body when $\mu = 0.4$, the contacts are not in force closure as indicated by the consistently labeled moment-labeling regions. As a result, some external forces cannot be balanced by contact forces

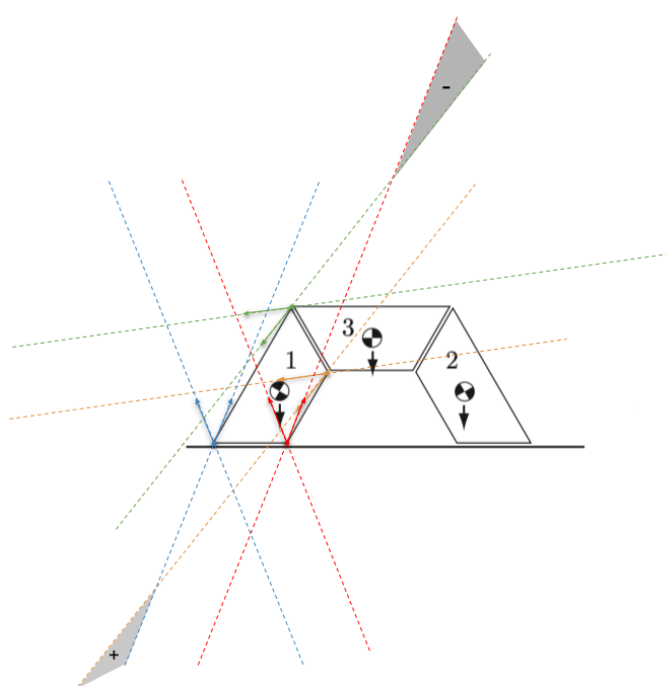

Figure 5

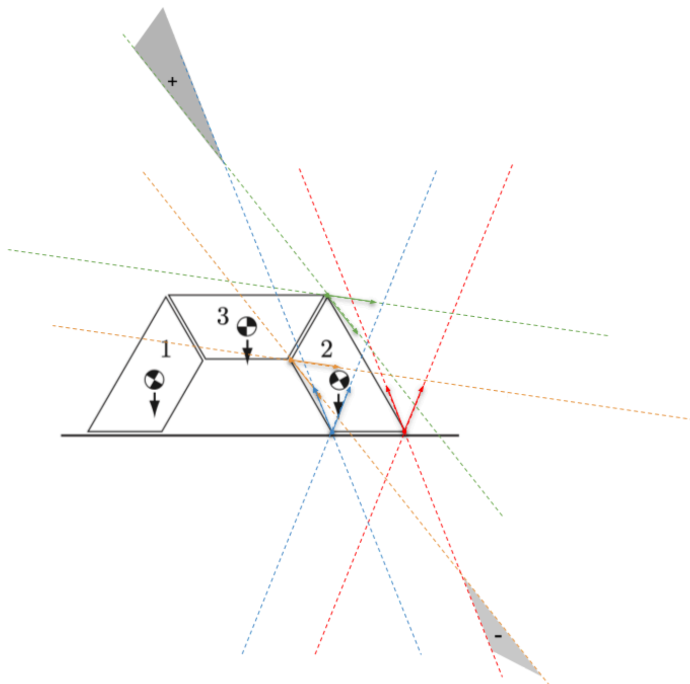

Figure 6

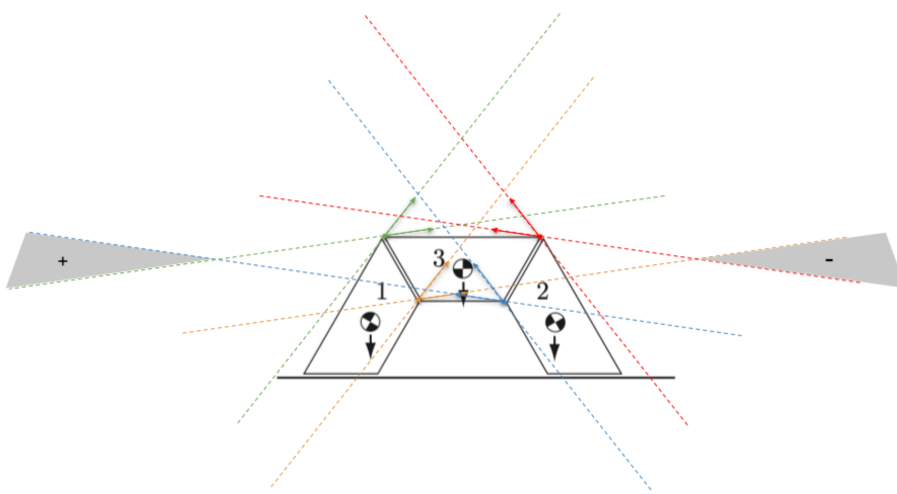

Figure 7